

Design and Simulation Based Studies of a Dual Band U-slot Patch Antenna for WLAN Application

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Abstract— Three different designs of a dual band patch antenna have been investigated in this paper. By using u-slot patch geometry dual band operation has been obtained. The patch antenna has been designed and simulated in CST Microwave Studio. By studying the antenna parametrically, it has been observed in CST Microwave Studio how the antenna characteristics vary with the variation of its different parameters. This observation has provided us sufficient insight to optimize the antenna parameters to meet the design requirements. The proposed designs of the antenna operate in 3.65 GHz band which is common to all three designs but the second operating band is different. By varying the u-slot broadness the second frequency band can be shifted from 5.25 GHz to 5.76 GHz. Moreover, it can be brought to 2.45 GHz band by carefully adjusting the parameters of the antenna. The gain and directivity in 3.65 GHz band of one of the three designs have been improved in other two designs. The gain has been improved from 3.73 dB to 6.6 dB and the directivity from 6.44 dBi to 7.55 dBi in 3.65 GHz band. The operating frequencies of the proposed designs fall in IEEE802.11a (5.20 GHz and 5.775 GHz), IEEE802.11b (2.45 GHz) and IEEE802.11y (3.675 GHz) standards which would allow WLAN operation. This patch antenna has been intended to be used in portable devices that demand miniaturized constituent parts.

Keywords— Patch antenna, u-slot, dual band, gain, directivity.

I. INTRODUCTION

The use of multiband antennas in portable devices like mobile phone, laptop, gaming console etc. is inevitable now-a-days. Due to the concept of miniaturization the size of these devices is shrinking rapidly. Consequently, the antennas must become smaller to fit inside them. This paper aims at presenting three miniaturized dual band u-slot patch antennas for WLAN application. The paper also confirms the technique of shifting resonance frequency by varying the u-slot broadness. Besides, the gain and directivity have been improved in two designs of the patch antenna. Apart from it, emphasis has been given to the miniaturization. At the end, the balance between the dimension and performance of the antenna has been carefully established by parametric study. For brevity, the results of parametric study have been omitted in this paper.

A simple patch antenna with basic rectangular patch geometry operates in a single frequency band. A patch antenna intended to operate at a centre resonance frequency f_r mounted on a substrate having dielectric constant ϵ_r would

have length L and width W of the patch as found from the following equations neglecting the fringing effect [1],

$$L = \frac{c}{2f_r\sqrt{\epsilon_r}} \quad (1)$$

$$W = \frac{c}{2f_r} \sqrt{\frac{2}{\epsilon_r + 1}} \quad (2)$$

where, c is the free-space speed of light.

A basic single band rectangular patch antenna can be modified into a multiband antenna by introducing slots in the patch. The shapes and position of the slot play an important role in determining the resonance frequency. The slot shapes like C, E, F, H, L, V, U and many others are well known. Sometimes, more than one slot of different shapes is used in the same patch for desired performance. The u-slot was introduced in a paper published in 1995 and the antenna showed wideband behaviour and was linearly polarized [2].

Although the initial investigations of u-slot patch antenna were based on air and foam substrate, later it was found that the antenna retained its wideband characteristic when material substrate was used instead of air and foam substrate [3], [4]. Initially, u-slot was regarded as a mean of achieving only wideband characteristic rather than multiband characteristic of the patch antenna. Interestingly, subsequent researches revealed that wideband characteristic can be modified to multiband characteristic by intelligent placement of u-slot, thereby perturbing the surface current flow in the patch [5], [6], [7]. Not only the band characteristic but also the polarization characteristic can be altered by manipulating u-slot. The antennas investigated in [8] and [9] exhibited circular polarization. Moreover, u-slot patch antenna with reconfigurable polarization has been reported recently [10]. In that paper, PIN diodes have been used to change the length of the u-slot arms, which alters the polarization state of the antenna and it can switch between linear and circular polarization. Besides, the antenna can also switch between right hand and left hand circular polarization. All these researches have unveiled different interesting aspects of u-slot patch antenna. It is now well established fact that by using u-slot patch geometry multiband characteristic can be obtained. Therefore, u-slot patch geometry is now regarded as a general template which can be customized to best suit the

requirements. The proposed antennas in this paper are three specific cases of the dual band characteristic of a general u-slot patch antenna template.

After intense parametric study and numerous simulations, three different designs of the dual band u-slot patch antenna named as Antenna A, B and C have been accomplished. Antenna A operates in 3.65 GHz (3.55–3.74 GHz) and 5.25 GHz (5.10–5.40 GHz) bands. Antenna B operates in 3.65 GHz (3.55–3.74 GHz) and 5.76 GHz (5.60–5.92 GHz) bands while Antenna C operates in 2.45 GHz (2.39–2.51 GHz) and 3.65 GHz (3.60–3.70 GHz) bands. All the operating bands of the designs have sufficient bandwidth to cover IEEE802.11a/b/y standards. The patch antennas proposed in this paper can find applications in portable devices supporting IEEE802.11a IEEE802.11b and IEEE802.11y standards. These standards are used for WLAN operation. Interestingly, the enhancement of IEEE802.11 standards using directional antennas has been reported recently [11], [12], [13].

II. ANTENNA DESIGN

A u-slot patch antenna on foam substrate was reported to exhibit dual band characteristic [14]. But using foam as the substrate of the patch antenna is not convenient at all. Therefore, material substrates are better choice. Another u-slot patch antenna on a microwave substrate exhibited broad-band characteristic [3]. The dimension of the ground plane in that paper was 120 mm × 100 mm, which is dimensionally large for mobile devices.

In this paper, the value of dielectric constant (ϵ_r) of the substrate in Antenna A and B is $\epsilon_r=2.2$ and in Antenna C $\epsilon_r=4.3$. The substrate thickness (h) is 4.5 mm. The substrate thickness (h) has been kept constant in all designs while the dielectric constant (ϵ_r) has been varied. The length and the width of the substrate are kept twice the length (L) and the width (W) of the patch. A ground plane having equal length and width of the substrate is used at the bottom of the antenna. Both the patch and the ground plane are made of electrically conducting material, preferably copper. The ground plane acts as a reflector of electromagnetic wave. The use of full ground plane ensures that the most of electromagnetic wave being impinged on the conducting metal ground plane is reflected back. The use of full ground plane creates a directional radiation pattern while the use of partial ground plane would have created an omnidirectional radiation pattern.

Coaxial probe feeding method has been used to feed the antenna. A coaxial cable with characteristic impedance of 50 Ω , which matches to the input impedance of the patch antenna, has been used. The use of coaxial probe introduces inductance which is cancelled by the capacitance introduced by u-slot. The u-slot patch antenna with dimensional variables is shown in Figure 1. The feeding mechanism of the proposed antenna is illustrated in Figure 2. The inner conductor of SMA connector is soldered to the patch and the outer conductor of SMA connector is soldered to the ground plane. The substrate is to be drilled carefully by a thin drilling bit to facilitate the insertion of the inner conductor of SMA connector. Drilling must be done in appropriate feeding position as indicated in Figure 1.

Otherwise antenna characteristics would change to a great extent.

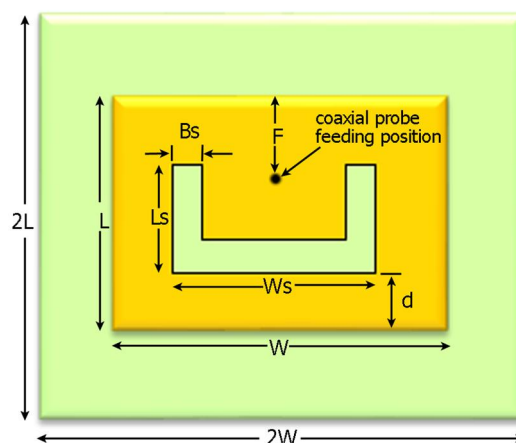


Figure 1. U-slot patch antenna with dimensional variables

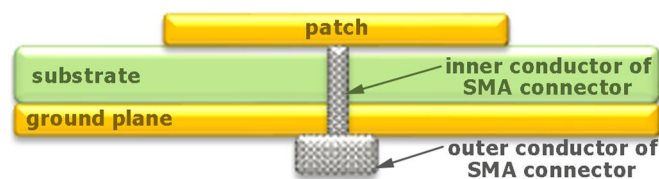


Figure 2. Coaxial probe feeding method

Table 1 lists the values of the parameters of the proposed patch antennas. All the dimensions are in mm except the dielectric constant (ϵ_r) which is unitless. Three different designs of the proposed u-slot patch antenna, named as Antenna A, B and C, have been investigated in this paper.

TABLE 1. PARAMETERS OF THE PROPOSED ANTENNAS

Designs	ϵ_r	h	d	L	W	L_s	W_s	B_s	F
Antenna A	2.2	4.5	4.81	21.5	26.5	8.5	12	1.25	8.5
Antenna B	2.2	4.5	4.81	21.5	26.5	8.5	12	2	8.5
Antenna C	4.3	4.5	5	24.5	26.5	8.5	12	1.2	8.5

It is worth mentioning here that the optimum values of the parameters of the proposed antennas have been obtained by intense parametric study, which is basically based on simulation. Most of the significant characteristics of the u-slot patch antenna have been firmly established by several experimental and simulation studies conducted by many researchers across the world. But no empirical formula of its design has been reported yet. Though empirical formulas for electrically thin substrates have been provided in a paper [4], design formulas for thick substrates are yet to be discovered. The proposed antennas in this paper have thick substrate with a thickness of 4.5 mm as seen from Table 1.

III. SIMULATION RESULTS

Simulations of the proposed antennas have been performed in CST Microwave Studio (version 2010), a commercial electromagnetic simulation software. Its simulation tool is based on FIT (Finite Integration Technique). Numerous simulations have been performed to confirm the results. Table 2 shows some important results of the different designs of the proposed antennas. For brevity, only relevant results have been presented in this paper.

TABLE 2. SIMULATION RESULTS OF THE PROPOSED ANTENNAS

Results	Antenna A	Antenna B	Antenna C
Resonance Frequencies (S_{11} dB)	3.65 GHz (-20.34 dB) 5.25 GHz (-23.37 dB)	3.65 GHz (-18.68 dB) 5.76 GHz (-13.17 dB)	2.45 GHz (-21.35dB) 3.65 GHz (-21.25 dB)
VSWR (Resonance Frequency)	1.21 (3.65 GHz) 1.13 (5.25 GHz)	1.26 (3.65 GHz) 1.56 (5.76 GHz)	1.18 (2.45 GHz) 1.18 (3.65 GHz)
Gain (Resonance Frequency)	6.62 dB (3.65 GHz) 6.34 dB (5.25 GHz)	6.66 dB (3.65 GHz) 6.44 dB (5.76 GHz)	5.28 dB (2.45 GHz) 3.73 dB (3.65 GHz)
Directivity (Resonance Frequency)	7.55 dBi (3.65 GHz) 7.49 dBi (5.25 GHz)	7.55 dBi (3.65 GHz) 7.61 dBi (5.76 GHz)	6.24 dBi (2.45 GHz) 6.44 dBi (3.65 GHz)

The return loss plot of the antennas is illustrated in Figure 3. Each sharp dip of the curve crossing -10 dB is representing a frequency band. Both Antenna A and C have good impedance matching because their return loss is less than -20 dB in all resonance frequencies.

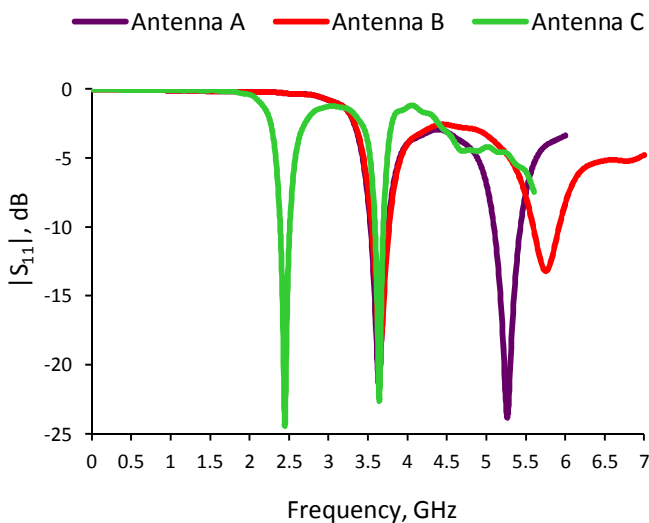


Figure 3. Return loss plot of the proposed antennas

The return loss in 3.65 GHz of Antenna B is -18.68 dB which also indicates a good impedance matching. Though the return loss curve of Antenna B merely crosses -10 dB in 5.76 GHz, the bandwidth of 5.76 GHz band is sufficient to accommodate the bandwidth required for IEEE802.11a standard. After calculation the bandwidth of 5.76 GHz band comes out to be $(5.92 - 5.60)$ GHz = 320 MHz.

By varying the u-slot broadness (B_s), the second resonance frequency of the u-slot patch antenna can be shifted from 5.25 GHz band to 5.76 GHz band. It can also be shifted to 2.45 GHz band as in the case of Antenna C. It should be noted that the first resonance frequency i.e. 3.65 GHz remains fixed in all three designs. The placement of u-slot disturbs the surface current in the patch, thus creating another band apart from the band created by the outer dimensions of the patch. The shift of resonance occurs due to the altered electrical path travelled by the surface current when the u-slot broadness (B_s) is varied.

The parameter S_{11} is also known as reflection coefficient, generally denoted by Γ . It is a complex number having magnitude and phase angle. Knowing the magnitude of Γ can be useful because the voltage standing wave ratio (VSWR) can be calculated using it by the following equation [15],

$$VSWR = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{1 + |S_{11}|}{1 - |S_{11}|} \quad (3)$$

Figure 4 illustrates the VSWR plot of the proposed antennas. As expected, Antenna B has the largest VSWR value 1.56 at 5.76 GHz because of inferior impedance matching as compared to the other two designs. The VSWR values of Antenna A and C are quite decent. Notice that the VSWR value at 3.65 GHz in all the designs does not show much variation. Its values at 3.65 GHz are 1.21, 1.26 and 1.18 in Antenna A, B and C respectively.

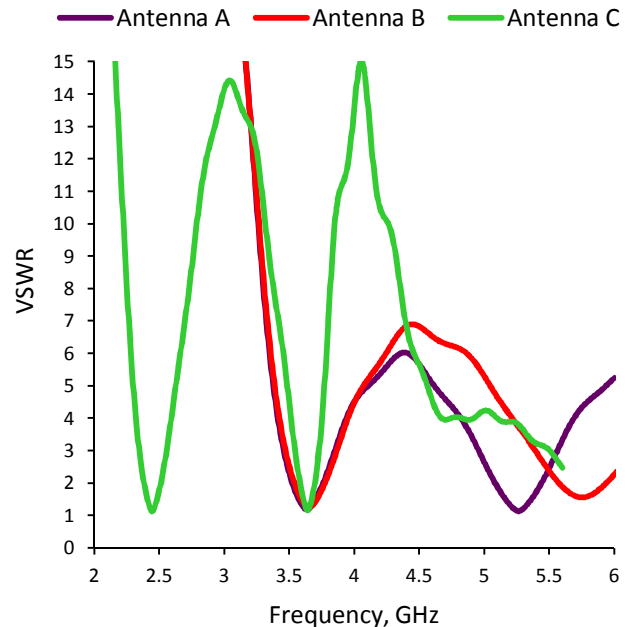


Figure 4. VSWR vs. frequency plot of the proposed antennas

The gain and directivity plots are shown in Figure 5 and Figure 6 respectively. The values of these two quantities are lower in Antenna C and they have been improved in Antenna A and B. Although the directivity has improved slightly from 6.44 dBi to 7.55 dBi, the gain has improved significantly from 3.73 dB to 6.6 dB in 3.65 GHz band. Four parameters, namely substrate dielectric constant (ϵ_r), distance of the u-slot from lower edge of the patch (d), length of the patch (L) and u-slot broadness (B_s) were varied to accomplish this improvement. Basically, lowering the substrate dielectric constant (ϵ_r) from 4.3 in Antenna C to 2.2 in Antenna A and B is a significant reason of this improvement. As expected, further significant improvement can be made by using air or foam substrate whose dielectric constant (ϵ_r) is 1. But as mentioned earlier, using air or foam substrate is not convenient for practical patch antenna design.

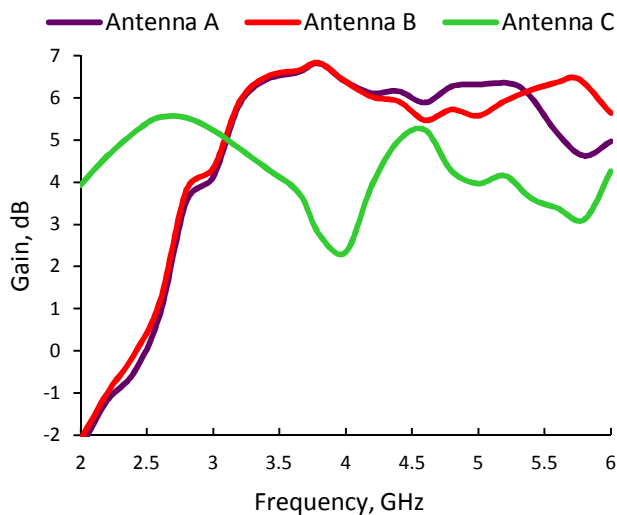


Figure 5. Gain vs. frequency plot of the proposed antennas

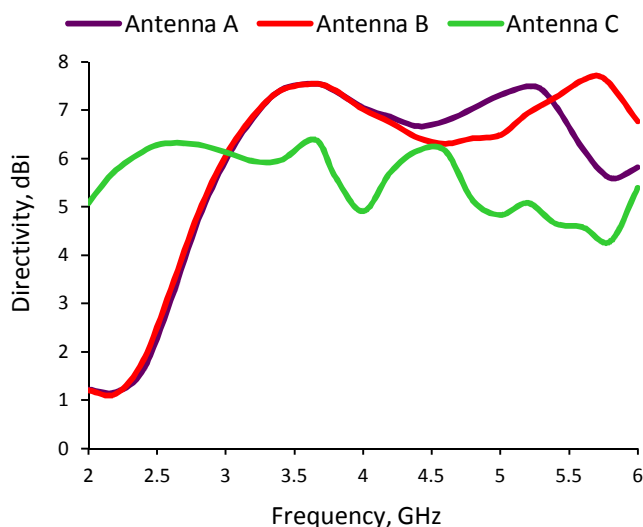


Figure 6. Directivity vs. frequency plot of the proposed antennas

The polar radiation patterns of Antenna A, B and C are shown in Figure 7, Figure 8 and Figure 9 respectively. These figures basically illustrate the power patterns plotted between θ , the elevation angle of spherical coordinate system and the power field. The polar patterns in these figures are shown in the elevation plane i.e. $\phi=90^\circ$, where ϕ is the azimuth angle of spherical coordinate system.

The proposed antennas are directional in nature and maximum radiation occurs at the top of the patch. The main lobe containing the maximum power is pointed above the patch. The precise direction of the main lobe has been provided below each plot. It has been indicated by the thick purple line in the polar plots. The 3 dB angular width, also known as half power beamwidth, in the elevation plane is mentioned under each polar plot. It is indicated as the angle between the two thin blue lines in the polar plots. As the directivity of the antenna increases, the half power beamwidth decreases. Therefore, the main lobe of the antenna becomes narrower. All the antennas have good radiation pattern as seen from the polar plots.

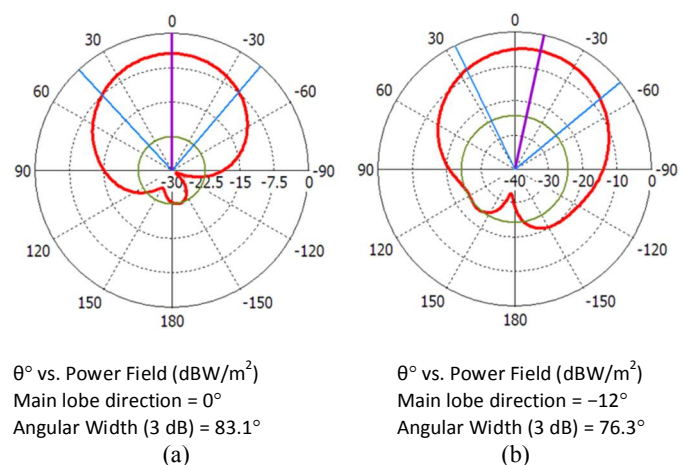


Figure 7. Polar radiation pattern of antenna A (a) 3.65 GHz (b) 5.25 GHz

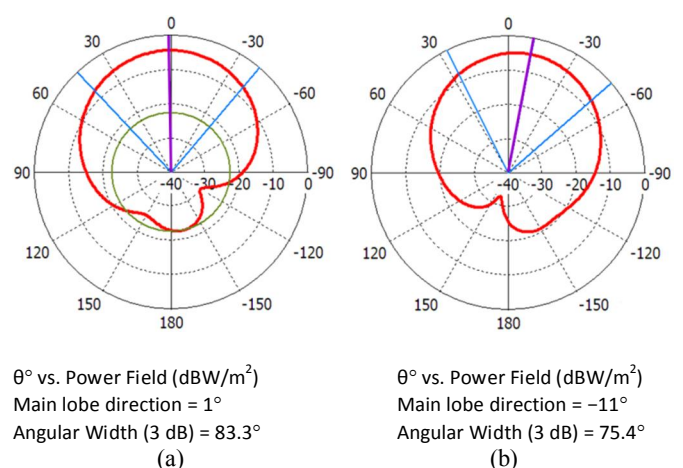


Figure 8. Polar radiation pattern of antenna B (a) 3.65 GHz (b) 5.76 GHz

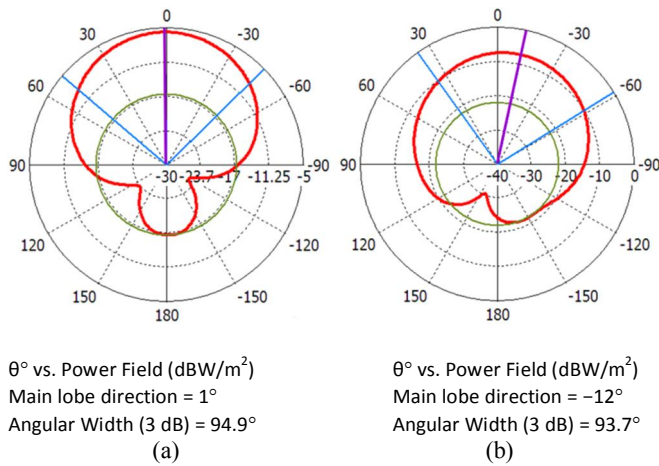


Figure 9. Polar radiation pattern of antenna C (a) 2.45 GHz (b) 3.65 GHz

Table 3 summarizes the frequency range and bandwidth required by IEEE802.11a/b/y standards. The operating frequency bands of the proposed antennas are also mentioned. The proposed antennas have sufficient impedance bandwidth to cover IEEE802.11a/b/y standards as found from Table 3.

TABLE 3. FREQUENCY BANDS OF THE PROPOSED ANTENNAS

Standards/ Antennas (Centre Resonance Frequency)	Frequency Range (GHz)	Bandwidth (MHz)
IEEE802.11a (5.20 GHz)	5.15 – 5.25	100
IEEE802.11a (5.775 GHz)	5.725 – 5.825	100
IEEE802.11b (2.45 GHz)	2.40 – 2.483	83
IEEE802.11y (3.675 GHz)	3.65 – 3.70	50
Antenna A (3.65 GHz & 5.25 GHz)	3.55 – 3.74	190
	5.10 – 5.40	300
Antenna B (3.65 GHz & 5.76 GHz)	3.55 – 3.74	190
	5.60 – 5.92	320
Antenna C (2.45 GHz & 3.65 GHz)	2.39 – 2.51	120
	3.60 – 3.70	100

IV. CONCLUDING REMARKS

Three different designs of a dual band u-slot patch antenna have been proposed for WLAN application. They are dimensionally small and have impressive performance

characteristics as pointed out in this paper. Even though this paper presents simulation based results, the use of CST Microwave Studio software for simulation ensures that there would not be large discrepancies between the simulated and measured results in case the proposed antennas are fabricated and measured. Being a powerful and commercial software, CST Microwave Studio is capable of delivering accurate simulation results. Besides, numerous simulations have been performed to confirm the results and the accuracy settings of the software have been carefully set up to minimize the approximation errors.

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